

You have $N \times C \times H \times W = 2 \times 2 \times 2 \times 2$ (That means batch size = 2, channel = 2, height = 2 and width = 2) Learned parameters: $\gamma = 3$, $\beta = 2$ for all positions. Use $\epsilon = 0$. Now, apply layer normalization for the following data:

	Channel 1		Channel 2	
Sample 1	1	1	3	3
	1	1	3	3
Sample 2	4	4	6	6
	4	4	6	6

Sample 1 – values: {1, 1, 1, 1, 3, 3, 3, 3}

- $\mu_{S1} = (1 \times 4 + 3 \times 4) / 8 = 16 / 8 = 2$
- $\sigma^2_{S1} = [(1-2)^2 \times 4 + (3-2)^2 \times 4] / 8 = (4 + 4) / 8 = 1$
- $\sigma_{S1} = 1$

Sample 2 – values: {4, 4, 4, 4, 6, 6, 6, 6}

- $\mu_{S2} = (4 \times 4 + 6 \times 4) / 8 = 40 / 8 = 5$
- $\sigma^2_{S2} = [(4-5)^2 \times 4 + (6-5)^2 \times 4] / 8 = (4 + 4) / 8 = 1$
- $\sigma_{S1} = 1$

Block	x	$\hat{x} = (x-\mu)/\sigma$	$\hat{y} = 3\hat{x} + 2$
S1, C1	1	$(1-2)/1 = -1$	-1
S1, C2	3	$(3-2)/1 = 1$	5
S2, C1	4	$(4-5)/1 = -1$	-1
S2, C2	6	$(6-5)/1 = 1$	5

More to do: solve the same problem for $\epsilon = 0.5$